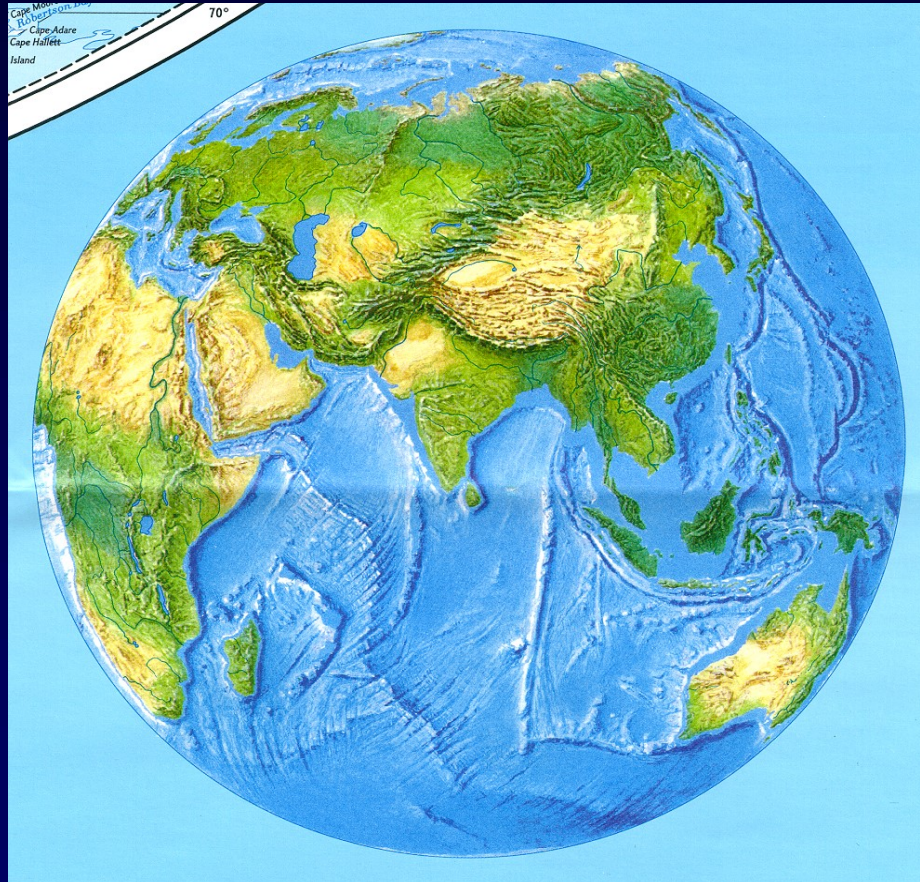


Keeling Curve



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Keeling Curve

- The Keeling Curve is a graph that represents the concentration of carbon dioxide (CO₂) in Earth's atmosphere since 1958.
- The Keeling Curve is named after Dr. Charles David Keeling, who started the CO₂ monitoring program at Mauna Loa Observatory in 1958 and supervised it until his death in 2005.
- Keeling's measurements showed the first significant evidence of rapidly increasing carbon dioxide (CO₂) levels in the atmosphere.
- Keeling began studying atmospheric carbon dioxide in 1958 by taking air samples and measuring the amount of CO₂ they contained. Over time he noticed a pattern. The air samples taken at night contained a higher concentration of CO₂ compared to samples taken during the day. He drew on his understanding of photosynthesis and plant respiration to explain this observation.

Keeling Curve.....continue

- His observation further revealed a steady increase in mean atmospheric CO₂ concentration from 313 parts per million by volume (ppmv) in March 1958 to 387.8 ppmv in July 2018.
- The data indicated a current increase of 2.48 ppmv CO₂ per year.
- This increase in atmospheric CO₂ is due to the combustion of fossil fuels, and has been accelerating in recent years.

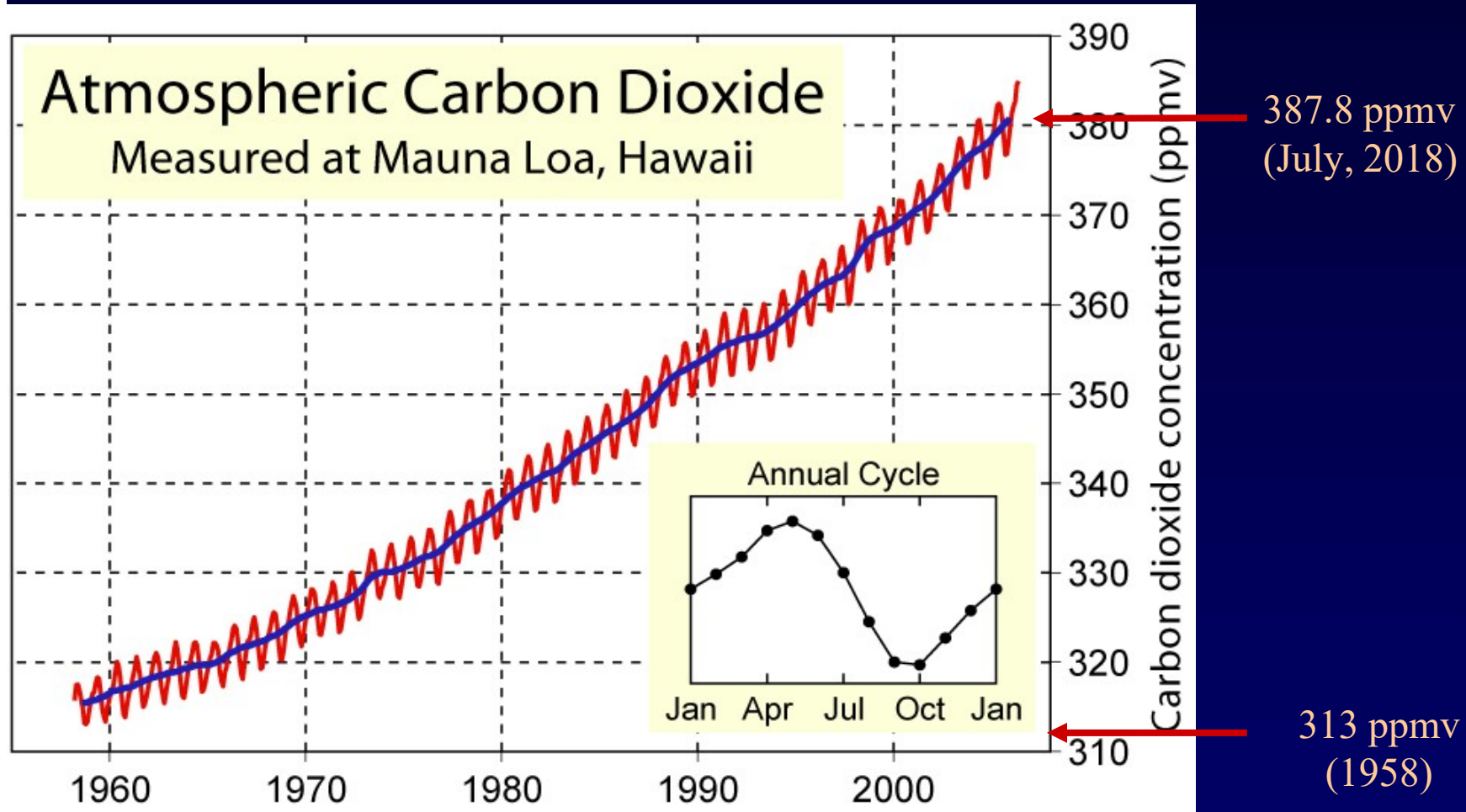
Keeling Curve.....continue

- Since CO₂ is a greenhouse gas, this has significant implications for global warming. Measurements of CO₂ concentration in ancient air bubbles trapped in polar ice cores show that mean atmospheric CO₂ concentration was between 275 and 285 ppmv during the Holocene epoch (9,000 BCE onwards), but started rising sharply at the beginning of the nineteenth century.
- The Keeling Curve also shows a cyclic variation of about 6 ppmv each year corresponding to the seasonal change in uptake of CO₂ by the world's land vegetation. Most of this vegetation is in the northern hemisphere where most of the land is located. From a maximum in May, the level decreases during the northern spring and summer as new plant growth takes CO₂ out of the atmosphere through photosynthesis. After reaching a minimum in September, the level rises again in the northern fall and winter as plants and leaves die off and decay, releasing CO₂ back into the atmosphere.

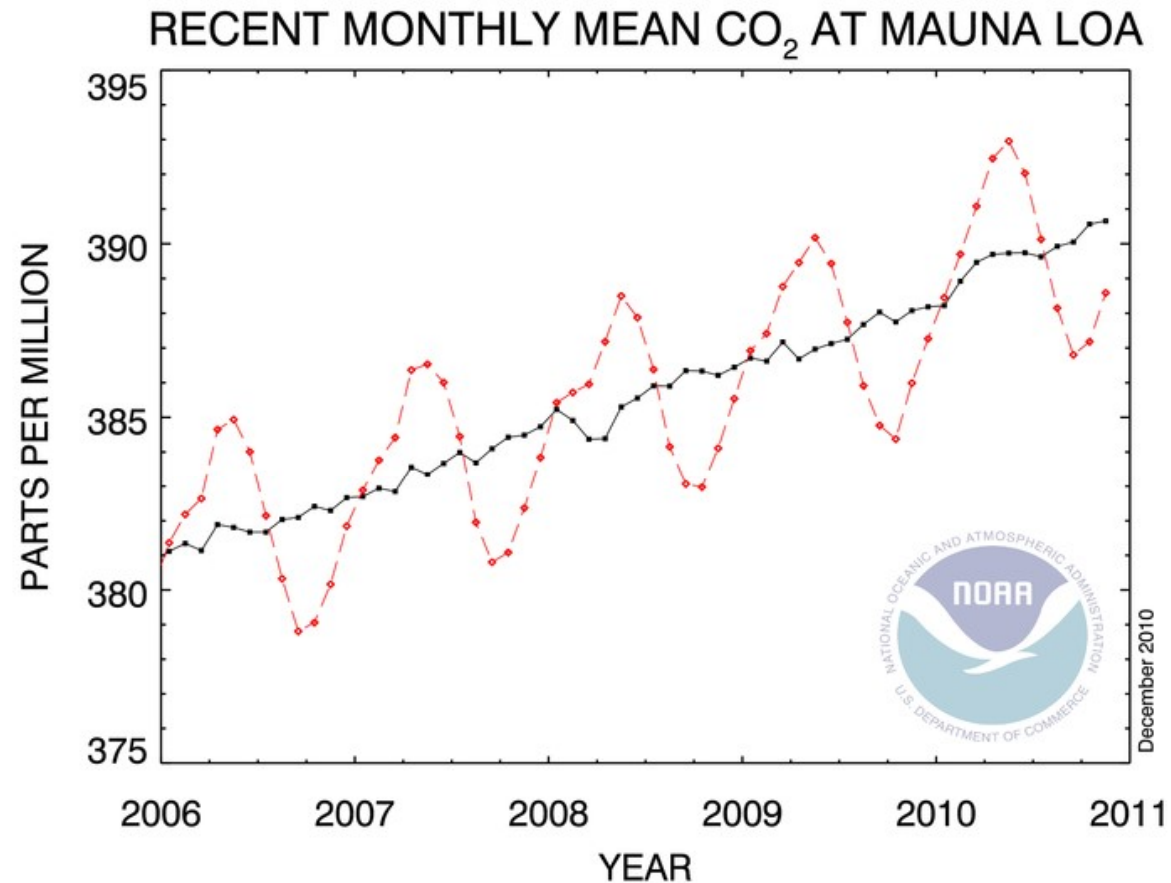
Keeling Curve.....continue

- Considering the significance of Keeling's findings, NOAA began monitoring CO₂ levels worldwide in the 1970s.
- Today, atmospheric CO₂ levels are monitored at about 100 sites around the globe through the Global Greenhouse Gas Reference Network.
- Measurements at many other isolated sites have confirmed the long-term trend shown by the Keeling Curve, although no sites have as long a record as Mauna Loa.
- In 2015, the Keeling Curve was designated a National Historic Chemical Landmark by the American Chemical Society.
- Commemorative plaques were installed at Mauna Loa Observatory and at the Scripps Institution of Oceanography at the University of California, San Diego in recognition of his efforts.

Keeling Curve



CO₂: the last few years



The red curve shows the actual data. The black curve is a smoothed fit. Rate of change: $10 \text{ ppm}/5 \text{ yr} = 2 \text{ ppm/yr}$



Thank You

Pillow lava at CIR observed during manned submersible dive